Notes on Bayesian Inference

Likelihoods • Priors • Posteriors • Prediction

Roadmap

- Likelihoods and MLE
- Priors (Beta)
- Bayes’ rule and the posterior
- MAP vs posterior mean
- Posterior predictive

Motivation: uncertainty-aware reasoning

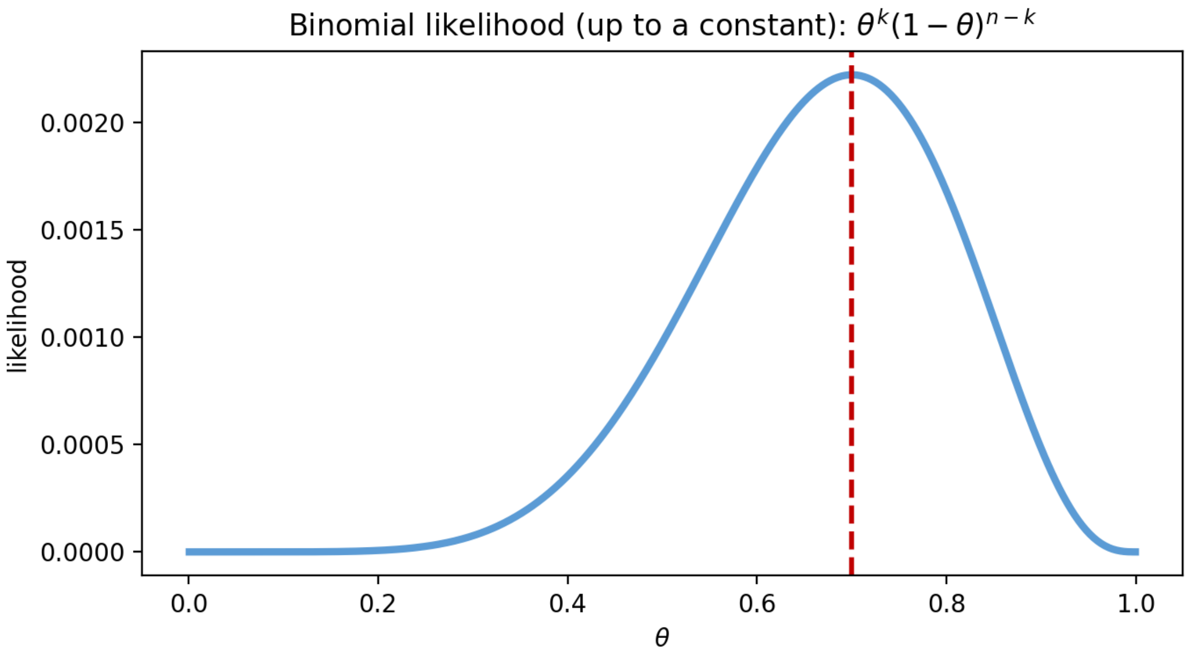
- MLE gives a single best θ, but ignores uncertainty.
- Bayesian inference returns a distribution over θ.
- This yields better-calibrated predictions when data are s

Coin flips as a Bernoulli model

Let $\theta = P(X=H)$.
Observe k heads in n flips.
Infer θ.

H H T T

Likelihood (Binomial)

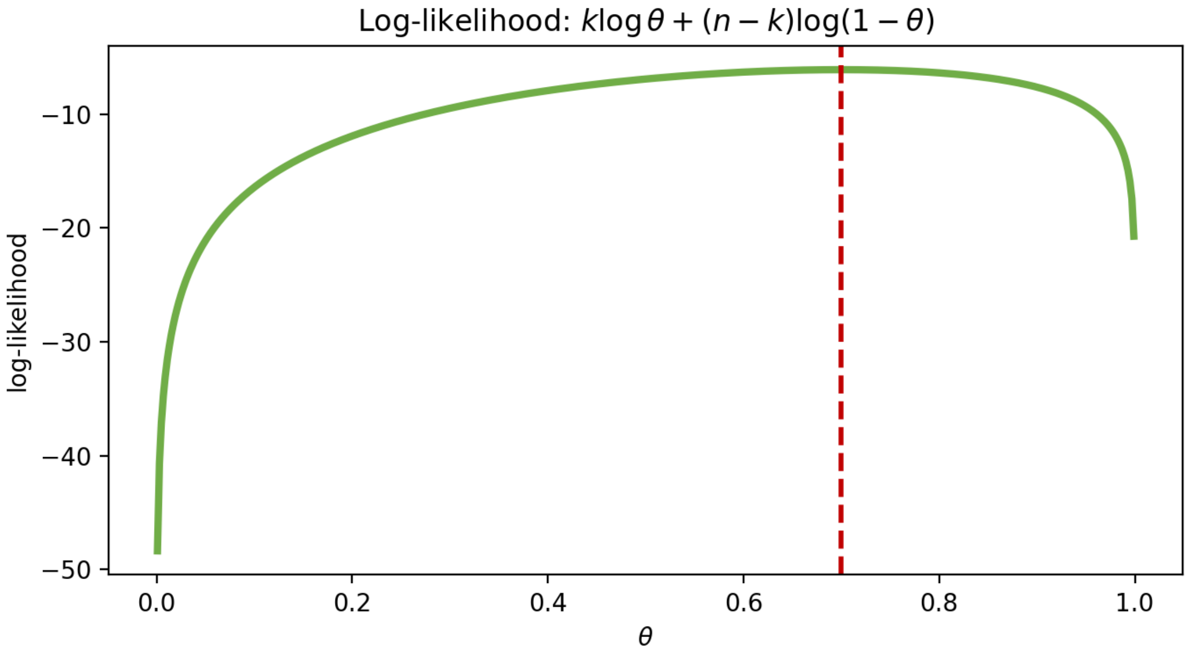


p(D|θ)=C(n,k) θ^k (

Treat as a function

Pick θ maximizing it

Log-likelihood



ℓ(θ)=k log θ + (n−k)

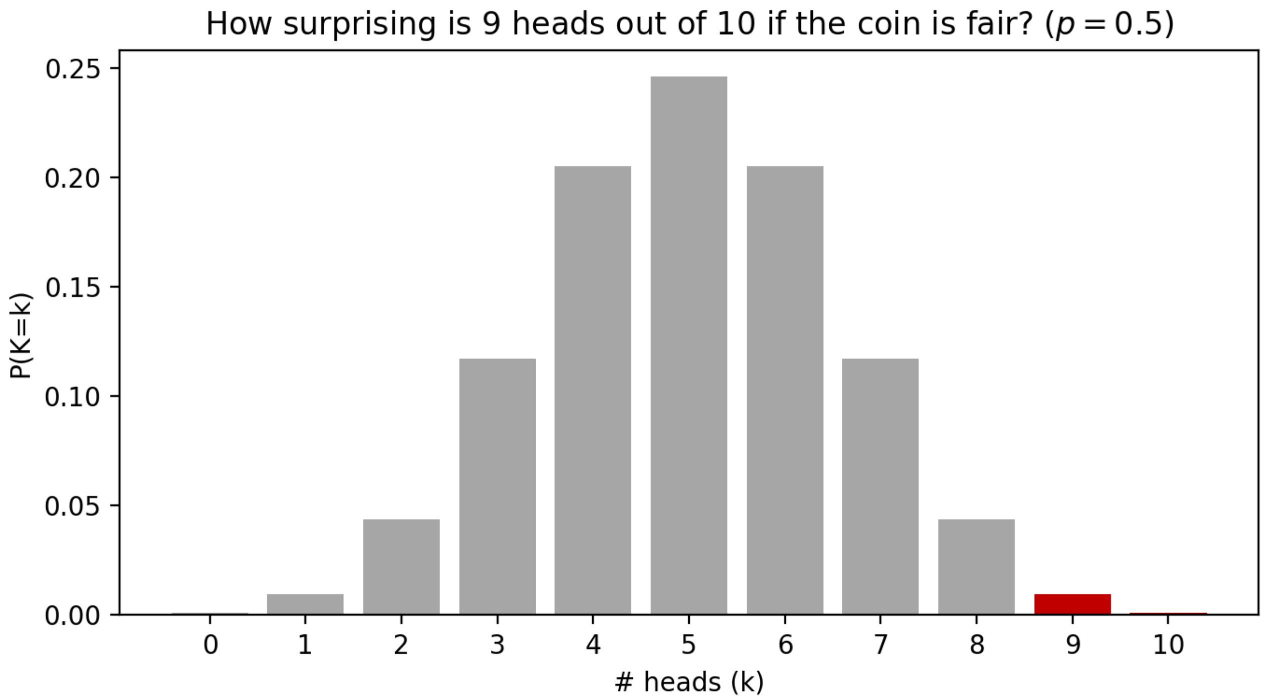
Derivative=0 ⇒ θ_M

MLE result

$$\theta\_MLE = k/n$$

(fraction of observed heads)

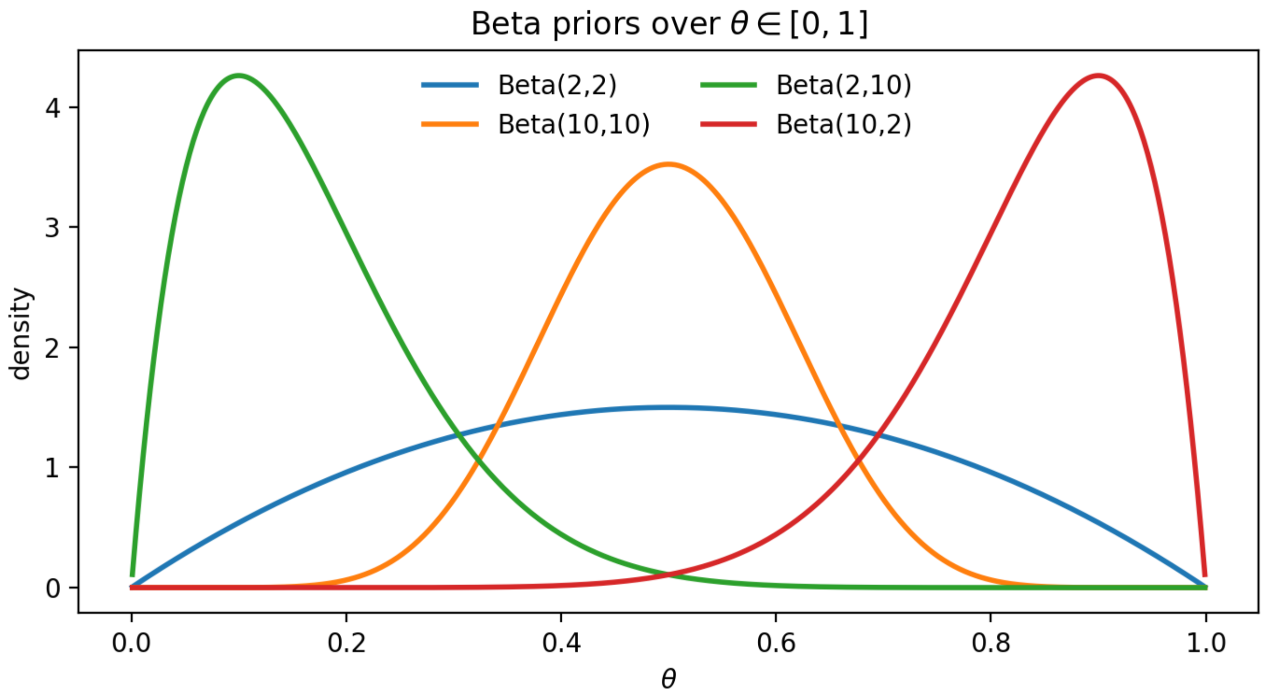
Why priors? Rare events happen



Even if θ=0.5, 9/

A prior lets us e

Beta priors over θ



Beta(α,β) is flexib

p(θ) ∝ θ^{α−1}(1

α,β control streng

Bayes' rule

Prior

$p(\theta)$

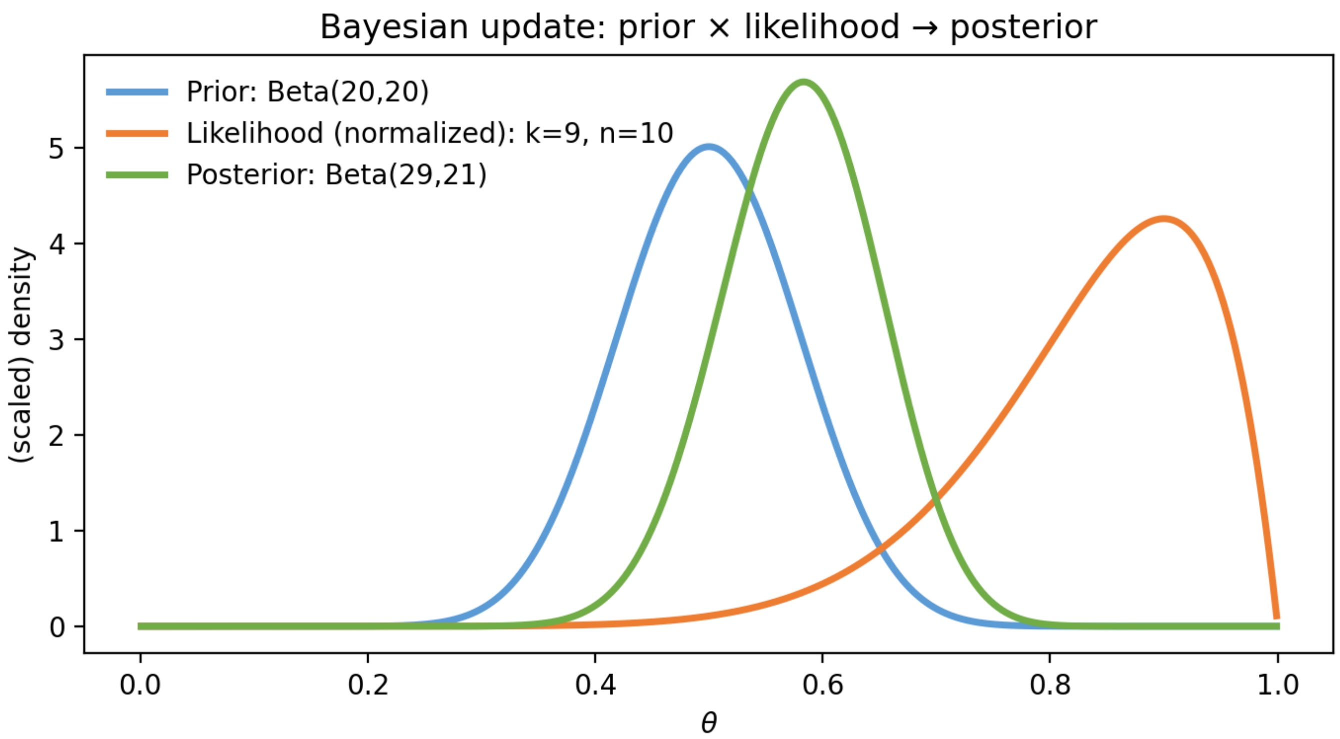
Likelihood

$p(D|\theta)$

$p(\theta|D)$

$$p(\theta|D) \propto p(D|\theta)\, p(\theta)$$

Conjugate update (Beta–Binomial)



If θ ~ Beta(α,

θ|D ~ Beta(α

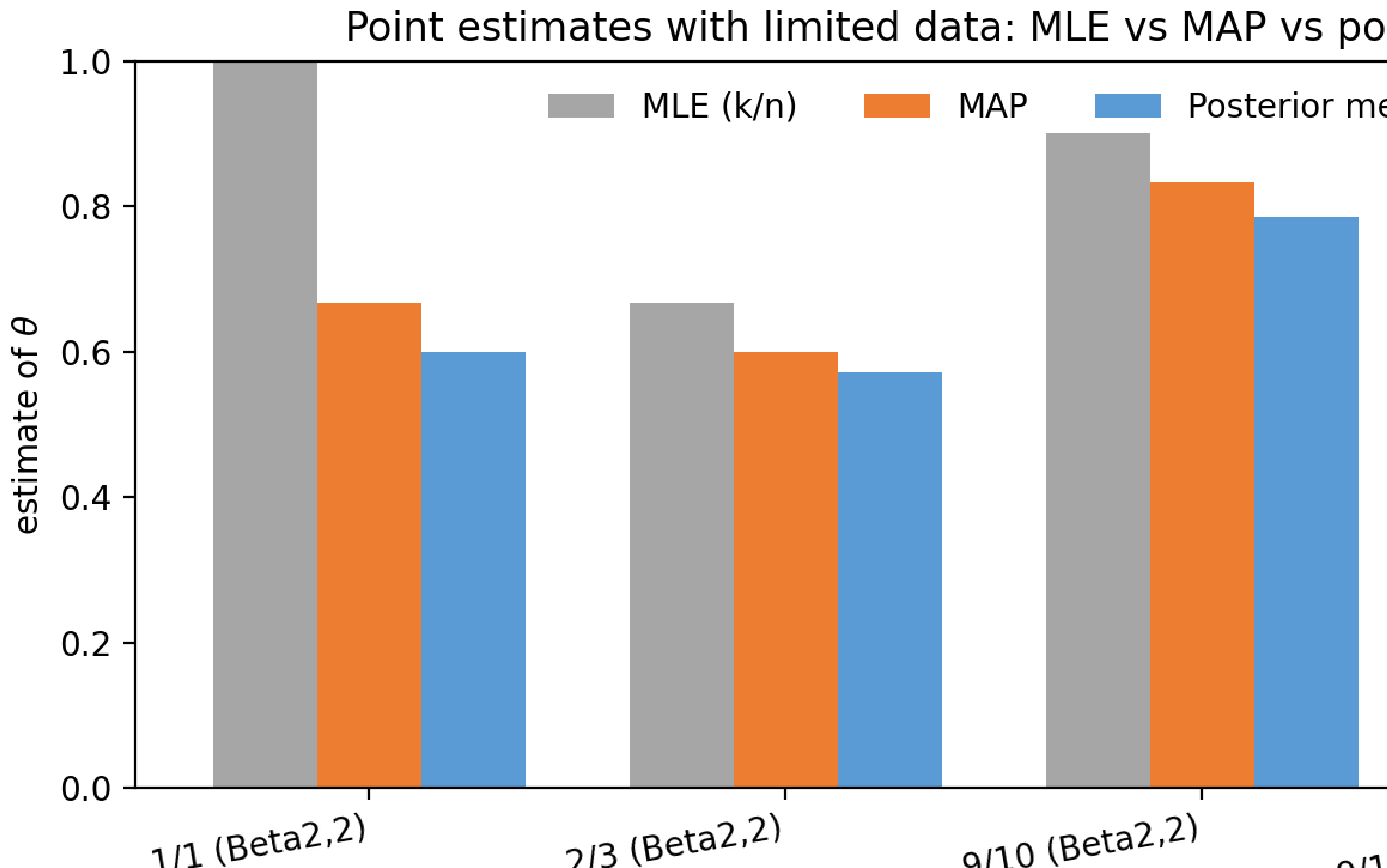
MAP vs posterior mean

$\theta\_MAP = (k+\alpha-1)/(n+\alpha+\beta-2)$

$E[\theta|D] = (k+\alpha)/(n+\alpha+\beta)$

$\theta\_MLE = k/n$

Point estimates: MLE vs MAP vs posterior mean

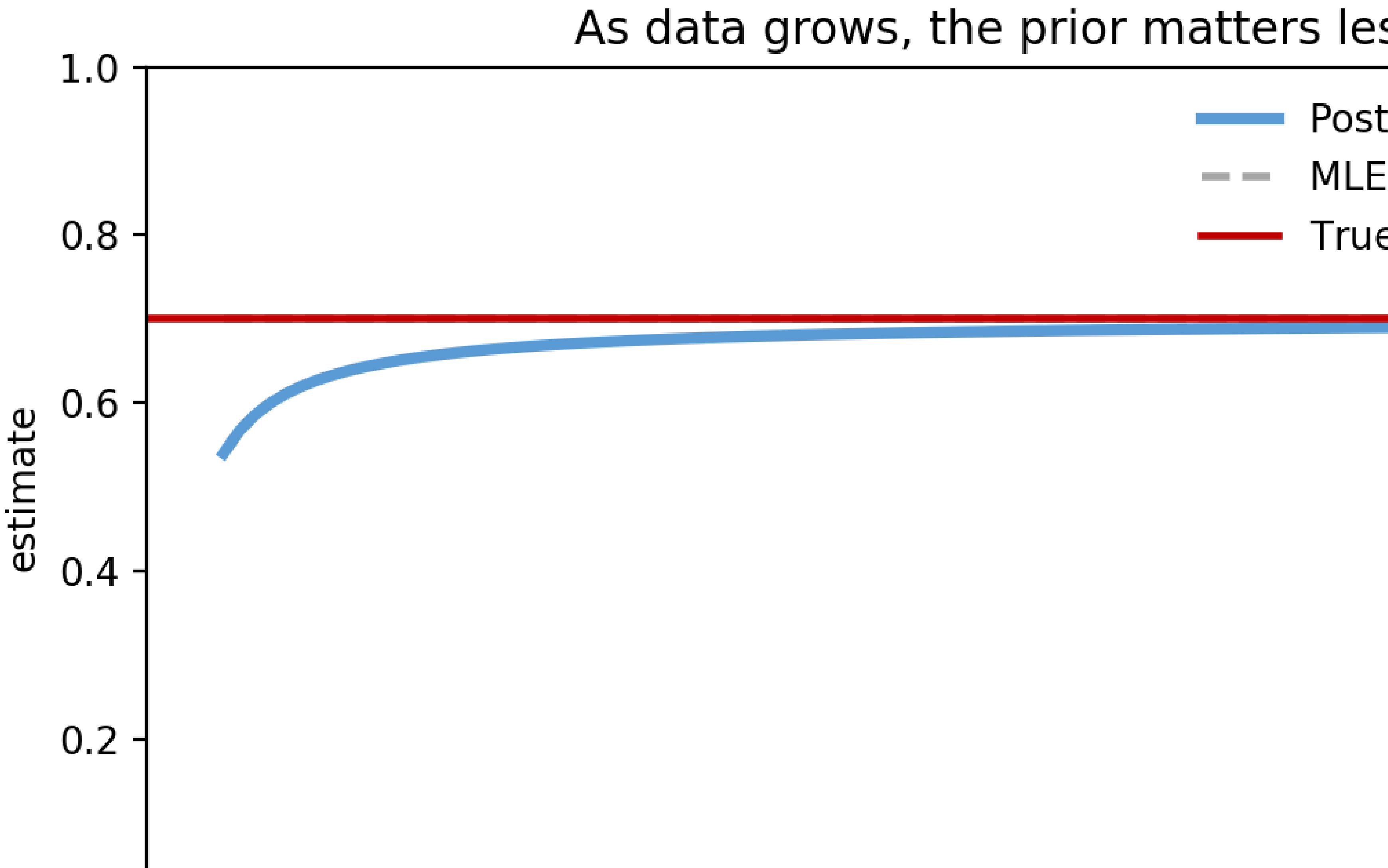


Posterior predictive

Predict next flip by averaging over θ:

$$P(X=H|D)=\int \theta\, p(\theta|D)\, d\theta = E[\theta|D]$$

More data → prior influence shrinks



Takeaways

- Bayesian inference models uncertainty via distributions over parameters.
- Posterior ∝ likelihood × prior (belief updating).
- Conjugacy gives closed-form updates for Beta–Binomial.
- Posterior predictive uses $E[\theta|D]$ for the next flip.

Interactive demo notebook

Included: Bayesian_Inference_Interactive_Demo.ipynb

Sliders for α, β, n, k update the prior, likelihood, poster